Program: **Diploma in Communication and Computer Networking**	
Course Code: **6328**	Course Title: **Advanced Computer Networking Lab**
Semester: **6**	Credits: **1.5**
Course Category: **Engineering Science**	
Periods per week: **3 (L:0 T:3 P:0)**	Periods per semester: **45**

Course Objectives:

- To understand the basics of networking, transmission control protocols
- To understand congestion control and traffic management
- To understand routing protocols
- To understand multimedia networking issues and approaches

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic concepts in computer networks	4322	Computer Networks	4

Course Outcomes:

On completion of the course, the student will be able to:

CO	Description	Duration (Hours)	Cognitive Level
CO1	Understand the fundamentals of networking and topologies	10	Understanding
CO2	Understand the basic principles of transmission media	10	Understanding
CO3	Understand the design issues of data link layer and error control	12	Applying
CO4	Understand the implementation and protocols used in network layer, transport layer and application layer	11	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	1		2		
CO2	3	1	2	1	2		
CO3	3	2	1	2			
CO4	3	2	1		2		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Understand the fundamentals of networking and topologies**		
M1.01	To understand the internet protocols-IPV4 and IPV6	3	Understanding
M1.02	To understand the transmission control protocols-TCP and UDP	3	Understanding
M1.03	To understand study packet switching networks X25, Frame relay	4	Understanding
Contents: **Network Basics:** The Internet Protocol(IPV4 and IPV6),Transmission Control Protocol(TCP),User Datagram Protocol(UDP),Packet Switching Networks-X.25,Frame Relay.			
CO2	**Understand the basic principles of transmission media**		
M2.01	To understand the congestion control mechanisms in computer networks	5	Understanding
M2.02	To study error control mechanisms	5	Understanding
	Series Test -I	1	

Contents: Congestion Control and Traffic management: Congestion Control Networks, Flow Control and Error Control, Asynchronous Transfer Mode(ATM)			
CO3	**Understand the design issues of data link layer and error control**		
M3.01	To study IPV4 addressing	2	Understanding
M3.02	To study routing algorithms	6	Applying
M3.03	To study IP subnetting	4	Understanding
Contents: **Routing:** Ip Addressing (IpV4), Adaptive and Non adaptive routing algorithms (Flooding, Shortest Path Routing, Distance Vector Routing), IP Subnetting(concepts only)			
CO4	**Understand the implementation and protocols used in network layer, transport layer and application layer**		
M 4.01	To understand audio and video compression techniques	6	Applying
M 4.02	To study audio and video streaming techniques	5	Understanding
	Series Test – II	1	
Contents: **Multimedia Networking Applications:** **Audio and Video compression techniques(preliminary ideas only),Streaming stored Audio /video, Streaming Live Audio/Video**			

Text / Reference

T/R	Book Title/Author
T1	William Stallings, “High Speed Networks and Internets – Performance and Quality of Service”, Pearson India 2005.
T2	Natalia Olifer Victor Olifer,” Computer Networks - Principles, Technologies and Protocols for Network Design”, - Wiley India (P) ltd. 2006.
T3	James F Kurose and Keith W Ross,” Computer Networking- A Top Down Approach Featuring Internet”, 3/e, Pearson Education.
T4	Kurose and Ross, “Computer Networks A systems approach”, Pearson Education.

R1	Behrouz A Forouzan, ”TCP/IP Protocol Suite”, Tata McGraw-Hill
R2	Behurouz A Forouzan, “Data Communications and Networking”, 4/e, McGraw-Hill.
R3	William Stallings, ”Data and Computer Communications “, 3/e,Pearson Education
R4	A. S. Tanenbaum, “Computer Networks”, 5/e, Prentice Hall India.

Online Resources

Sl.No	Website Link
1	http://www.edrawsoft.com